COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 -Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
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ACTIVITY NO. 4.1 **Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Implement a stack using the C++ STL (std::stack).

main.cpp mainb.cpp

ILO A > main.cpp > main()

```cpp
#include <iostream>
#include <stack>

int main(){
    std::stack<int> stack1;
    std::cout<<"Testing the stack STL:"<<std::endl;

    //isEmpty
    std::cout<<"Is the stack empty? " <<stack1.empty()<<std::endl;
    //push
    stack1.push(2);
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    stack1.push(1);
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    stack1.push(3);
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    stack1.push(4);
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    //pop
    stack1.pop();
    std::cout<<"The size of the stack is "<<stack1.size()<<std::endl;
    stack1.top();
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    //isEmpty
    std::cout<<"Is the stack empty? " <<stack1.empty()<<std::endl;
    //emplace
    stack1.emplace(6);
    std::cout<<"The top of the stack is "<< stack1.top()<<std::endl;
    //before pop
    std::cout<<"The size of the stack is "<<stack1.size()<<std::endl;
    stack1.pop();
    //after pop
    std::cout<<"The size of the stack is "<<stack1.size()<<std::endl;
    //swap
    std::stack<int> stack2;
    //push in stack2
    stack2.push(9);
    std::cout<<"The top of the stack is "<< stack2.top()<<std::endl;
    stack2.push(8);
    std::cout<<"The top of the stack is "<< stack2.top()<<std::endl;
    stack2.push(7);
    std::cout<<"The top of the stack is "<< stack2.top()<<std::endl;
    stack2.push(6);
    std::cout<<"The top of the stack is "<< stack2.top()<<std::endl;
    //swapping
    stack1.swap(stack2);
    //results
    std::cout<<"Stack 1: " << stack1.size() << std::endl;
    std::cout<<"Stack 2: " << stack2.size() << std::endl;

}
```

```
Testing the stack STL:
Is the stack empty? 1
The top of the stack is 2
The top of the stack is 1
The top of the stack is 3
The top of the stack is 4
The size of the stack is 3
The top of the stack is 3
Is the stack empty? 0
The top of the stack is 6
The size of the stack is 4
The size of the stack is 3
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
The top of the stack is 6
Stack 1: 4
Stack 2: 3
PS C:\Users\TIPQC\Desktop\VILLARUEL CPE21S2\HOA 4.1\ILO A>
```

Observation: For ILO A, I was able to apply all functions required in stack, including push, pop, top, emplace and

swap. Pushing and Emplacing allows you to add a value in the stack, wherein the value you add the latest will serve as the top of your stack. While popping performs exactly the opposite, in which it removes the latest value you added. Lastly, swapping allows you to swap the value of two different stacks.

ILO B.1: Implement a stack using a Static Array.



```cpp
#include<iostream>

//global direction
#define maxCap 10
int choice;
int stackArr[maxCap];
int top = -1, newData;
//prototype functions
void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
void displayStack();


int main(){

//main driver

    while (true)
    {
        std::cout<<"Stack operations \n";
        std::cout<<"1. Push 2. Pop 3. Top 4. isEmpty 5. isFull 6. Display"<<std::endl;
        std::cout<<"========================================"<<std::endl;
        std::cin >> choice;

        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout<<"Is stack empty? "<< isEmpty()<<std::endl;
            break;
            case 5: std::cout<<"Is stack full? "<< isFull()<<std::endl;
            break;
            case 6: displayStack();
            break;
            default: std::cout<<"Invalid choice"<<std::endl;
            break;

        }
    }
}

//function definition
```



```cpp
//function definition
bool isEmpty() {
    //verifying if the stack is empty
    if(top == -1) return true;
    return false;
}

bool isFull() {
    //verifying if the stack is full
    if(top == maxCap - 1) return true;
    return false;
}

void push() {
    if(isFull()){
        std::cout<<"Stack overflow"<<std::endl;
        return;
    }
    //pushing
    std::cout<<"Enter a new value: "<<std::endl;
    std::cin>> newData;
    //insert
    stackArr[++top] = newData;
}
void pop() {
    if(isEmpty){
        std::cout<<"Stack underflow"<<std::endl;
    }
    std::cout<<"Popping: "<<stackArr[top]<<std::endl;
    top--;
}

void Top(){
    if(isEmpty){
        std::cout<<"The stack is empty"<<std::endl;
        return;
    }
    //check the top value
    std::cout<<"Top element: "<<stackArr[top]<<std::endl;
}

void displayStack(){
    if(isEmpty()){
        std::cout<<"There is nothing to display"<<std::endl;
        return;
    }
    std::cout<<"All stack elements: "<<std::endl;
    for(int i = top; i<= -1; i--){
        std::cout << stackArr[i] << std::endl;
    }


}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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